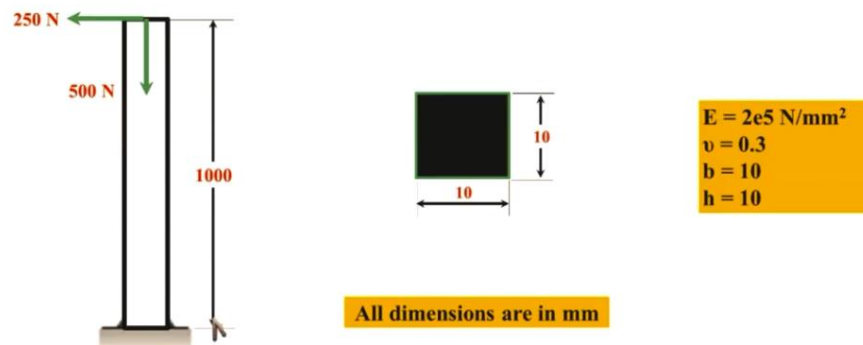


Practical 4

Title – Beam Element – Non-Linear Buckling Analysis

Sample Example

NON-LINEAR BUCKLING ANALYSIS OF A SLENDER COLUMN BY USING ANSYS



Procedure

Step 1 Preferences

Open Ansys APDL – Preferences – Structural – OK

Step 2 Pre-processor

Pre-processor – Element Type – Add – Beam – 3 nodes 189 - OK - Close

Step 3 Material

Material Properties – Material Models – Structural – Linear – Elastic –
Isotropic – Enter Young's Modulus – Enter Poisson's Ratio - OK

Step 4 Sections

Sections – Beam - Common Sections - Enter B & H 10 – Preview - OK

Step 5 Modeling

Modeling – Create –Key point 1 –In Active CS - 0,0 – Apply – Key point 2 – 0, 1000 – OK

Modeling – Create – Lines – Straight Lines – OK

Step 6 Meshing

Meshing – Size Controls – Manual Size – Lines – All Lines – No of Element Division 100 – Ok

Mesh – Lines – Pick All

Step 7 Loads

Loads – Define Loads –Apply – Structural – Displacement – Key point 1 – Ok – All DOF - Ok

Loads – Define Loads –Apply – Structural – Force– Key points – FX - -250 – Apply – FY - -500 - Ok

Step 8 Solution

Solution – Analysis Type – New Analysis – Static - OK

Solution – Analysis Type – Solution Controls – Large Displacement Static – No of Sub steps 20 – Max number of sub step 1000 - min number of sub steps 1

Frequency – Write every substep

Solution – Analysis Type – Solution Controls – Nonlinear – line search – on – maximum number of Iterations- 1000 - OK

Step 9 Solution

Solution – Solve –Current LS - OK

Step 10 General Results

General Post Process – Read Results – By Pic – Close

General Post Process – Plot Results – Deformed Shape – Deformed + Undeformed - OK

Plot Controls – Style – Size and Shape – Display of Shape - ON

General Post Process – Plot Results – Deformed Shape -

To See Animation

Plot Control – Animate – Deformed Shape – Deformed + Undeformed - OK

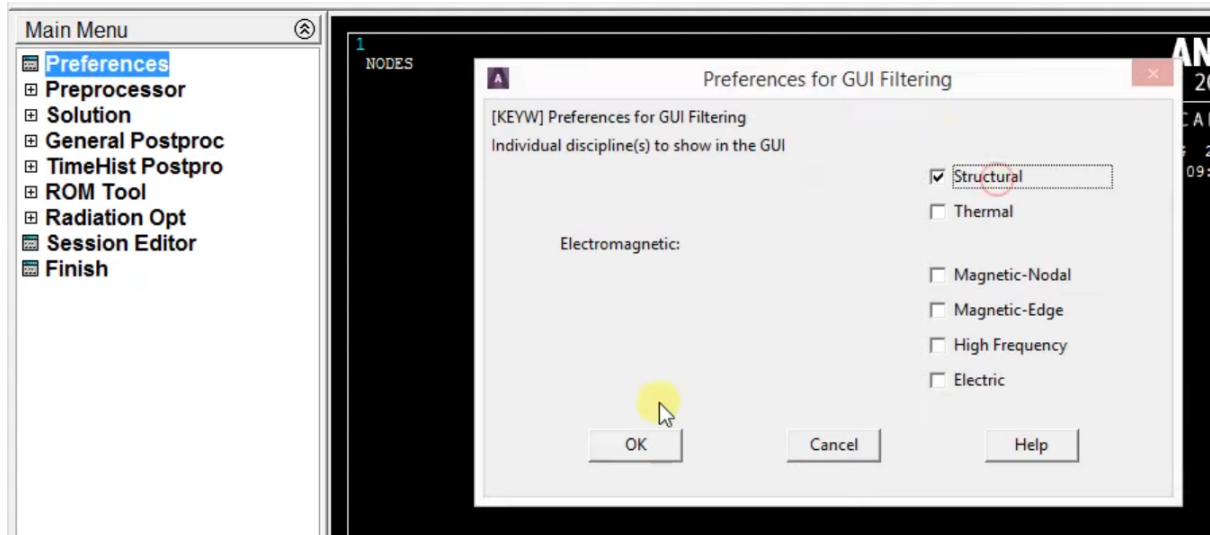
General Post Process – Plot Results – Contour Plot – Nodal Solution – DOF Solution – X Component of displacement – ok

General Post Process – Plot Results – Contour Plot – Nodal Solution – DOF

Solution –displacement vector sum - ok

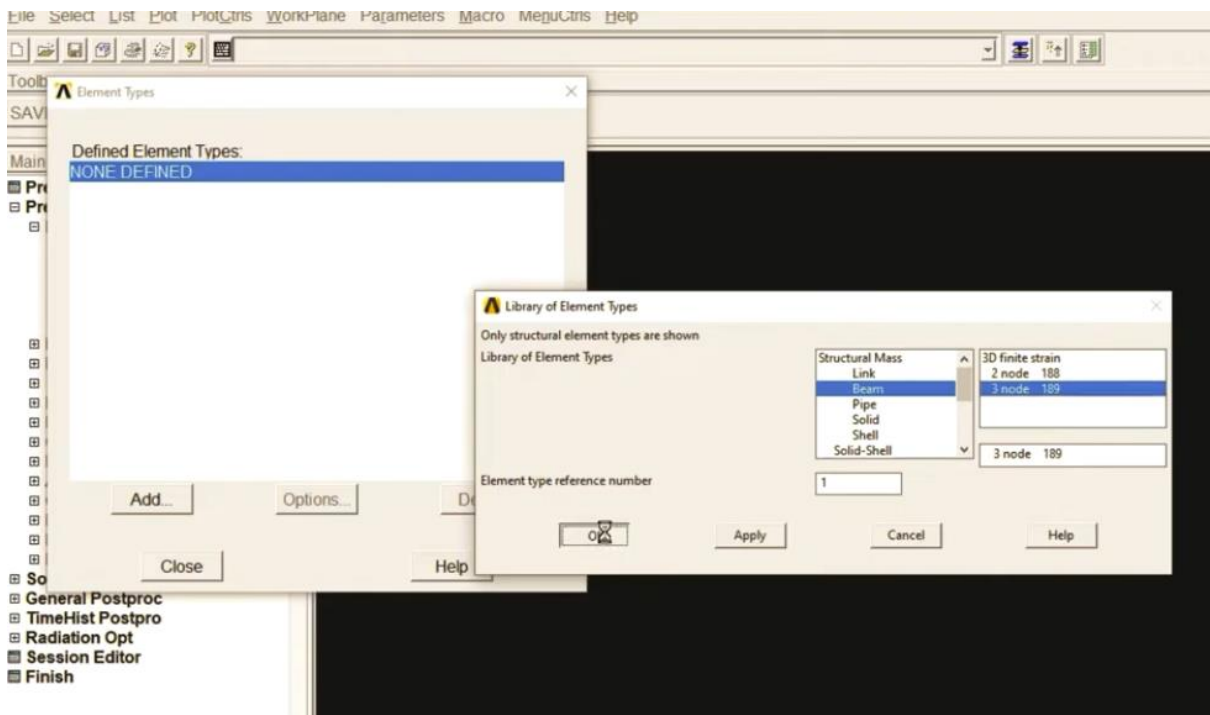
Step 1 Preferences

Open Ansys APDL – Preferences – Structural – OK



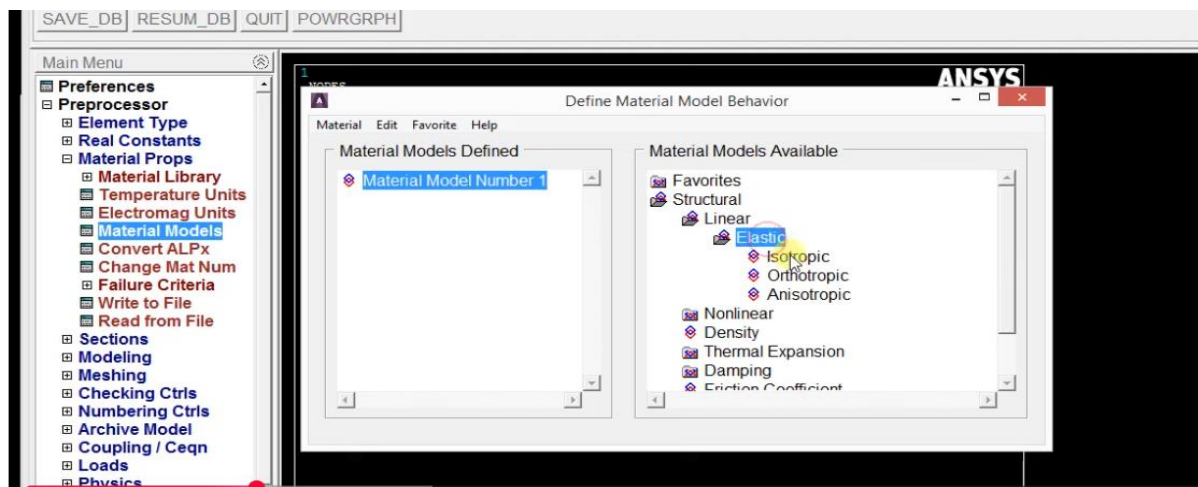
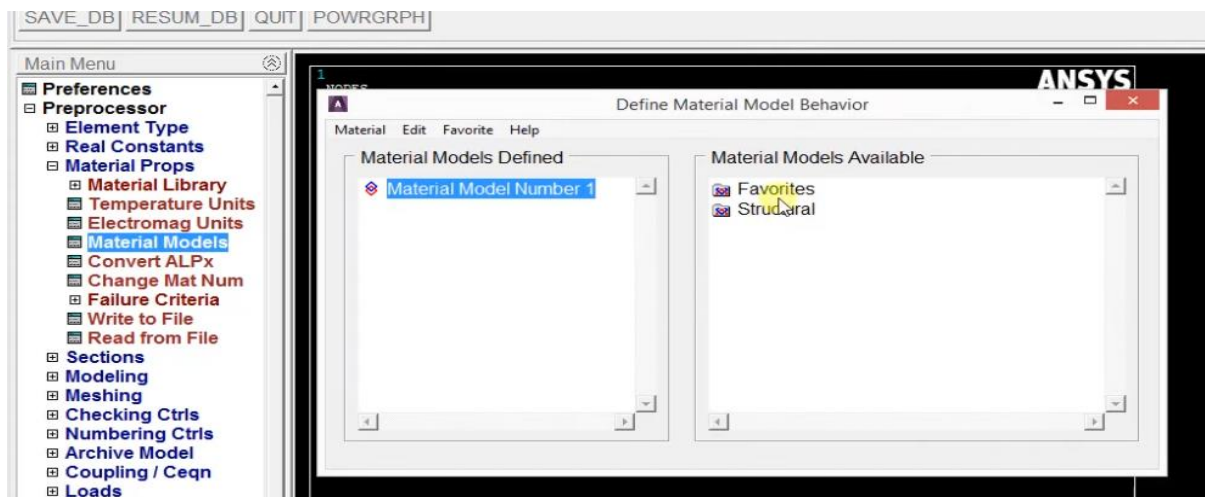
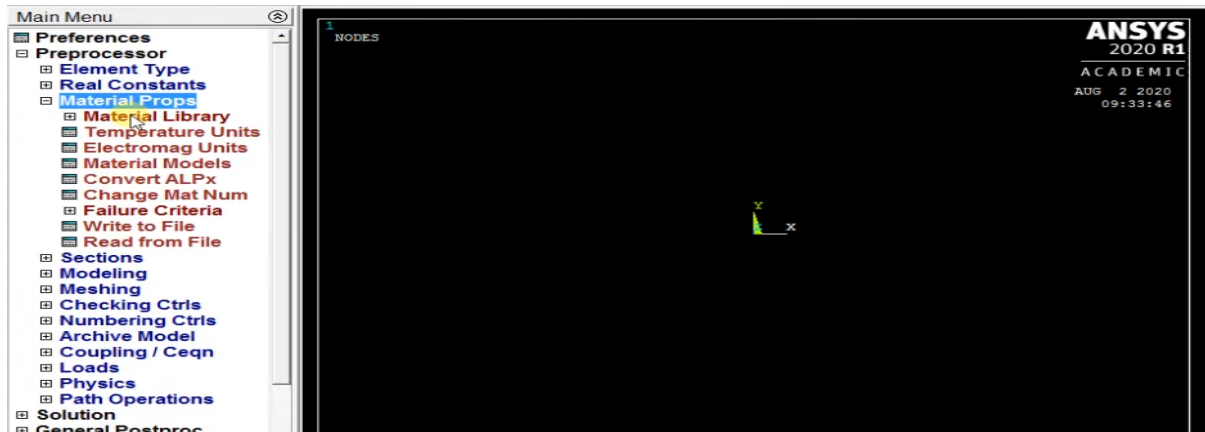
Step 2 Pre-processor

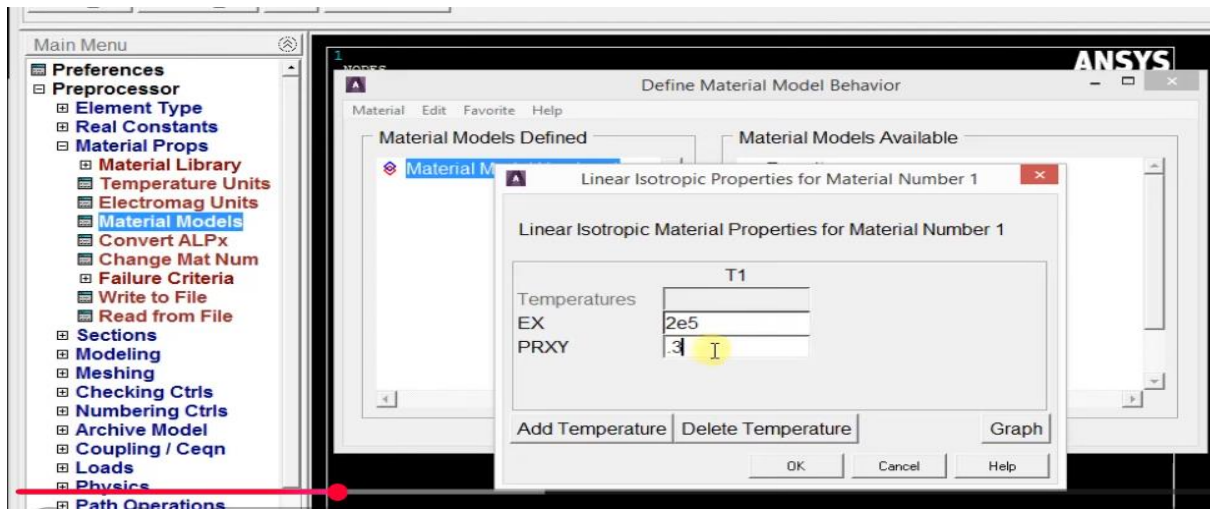
Pre-processor – Element Type – Add – Beam – 3 nodes 189 - OK - Close



Step 3 Material

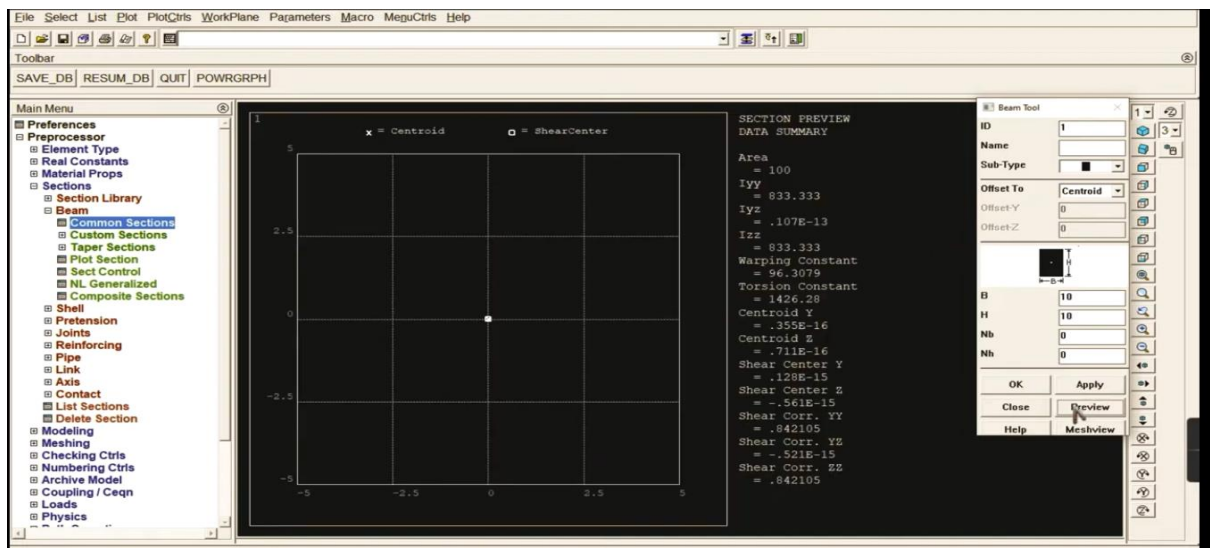
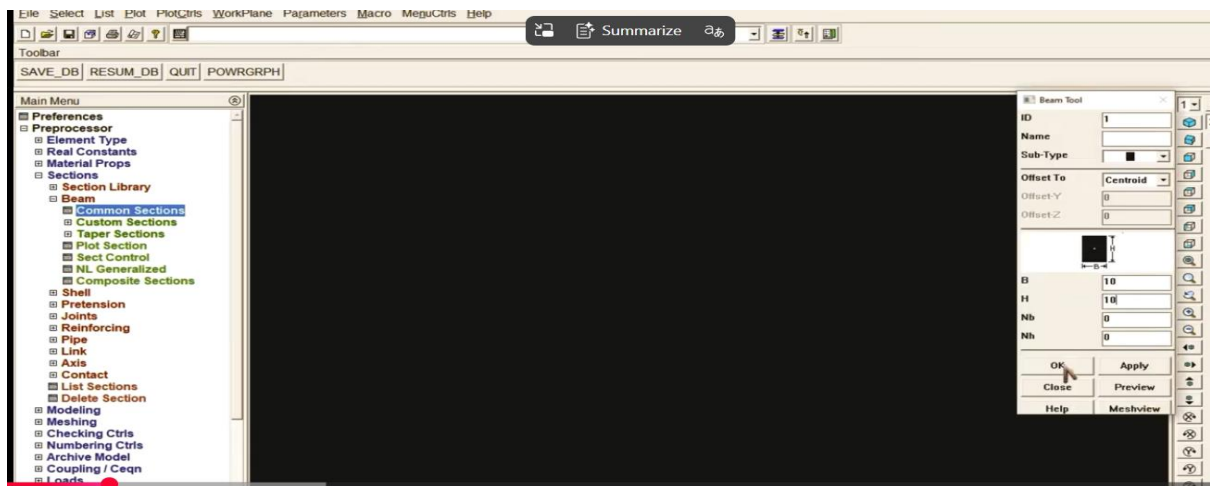
Material Properties – Material Models – Structural – Linear – Elastic – Isotropic – Enter Young's Modulus – Enter Poisson's Ratio - OK





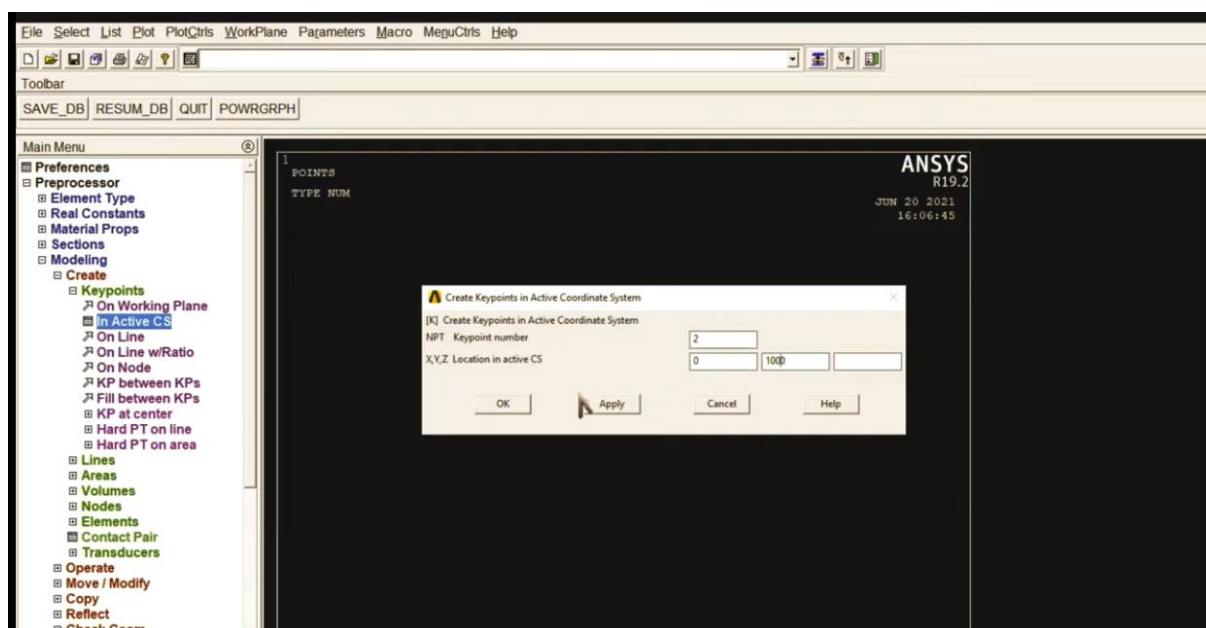
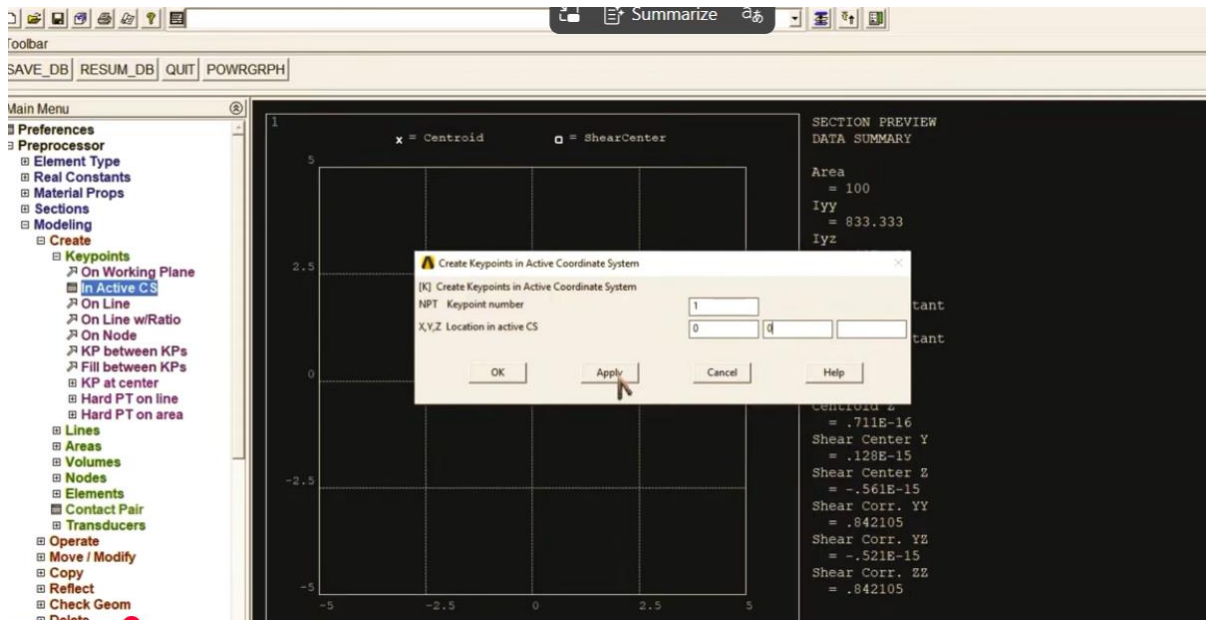
Step 4 Sections

Sections – Beam – Common Sections – Enter B & H 10 – Preview - OK

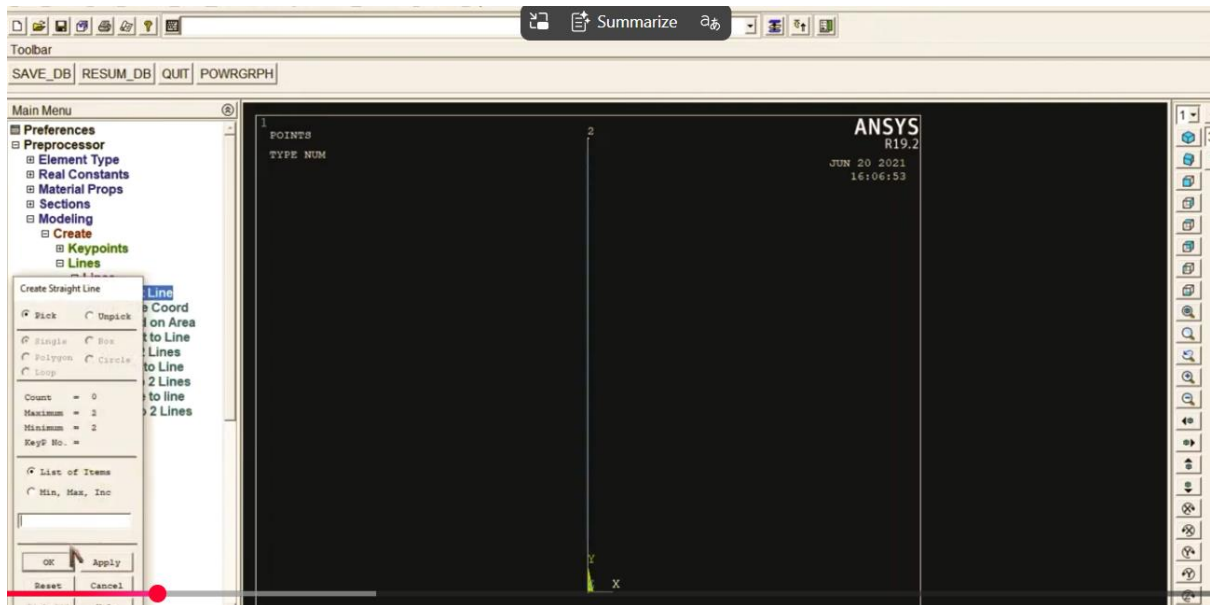


Step 5 Modeling

Modeling – Create –Key point 1 –In Active CS - 0,0 – Apply – Key point 2 – 0, 1000 - OK

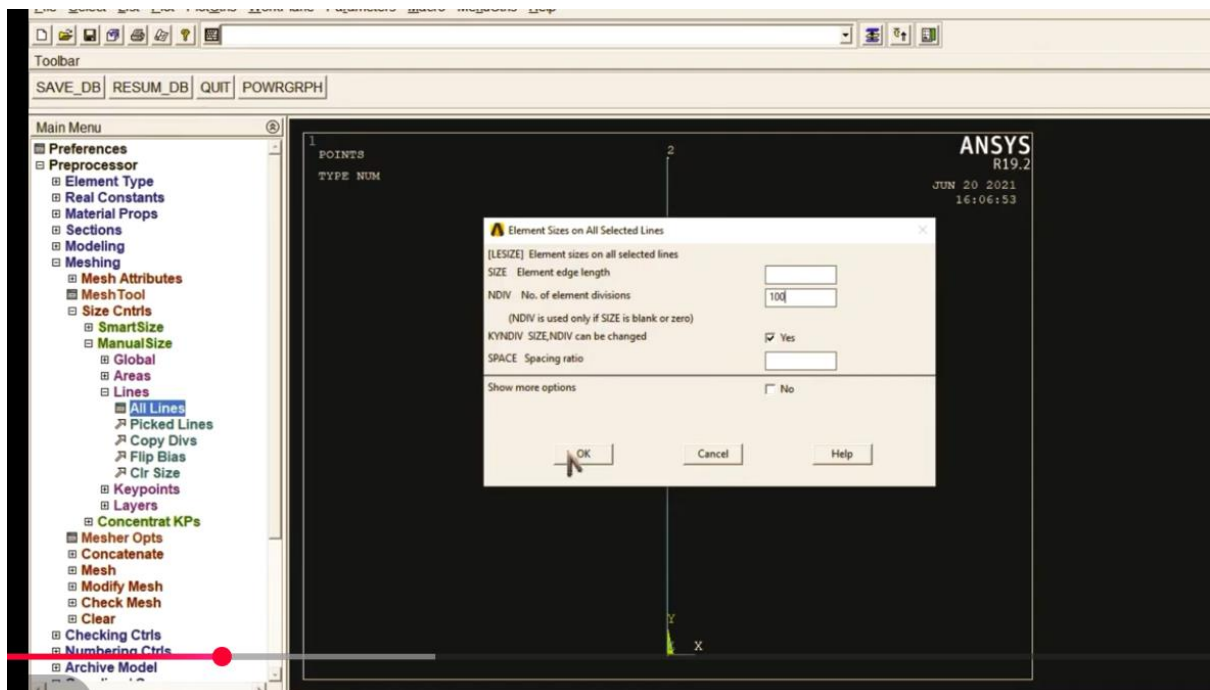


Modeling – Create – Lines – Straight Lines – OK

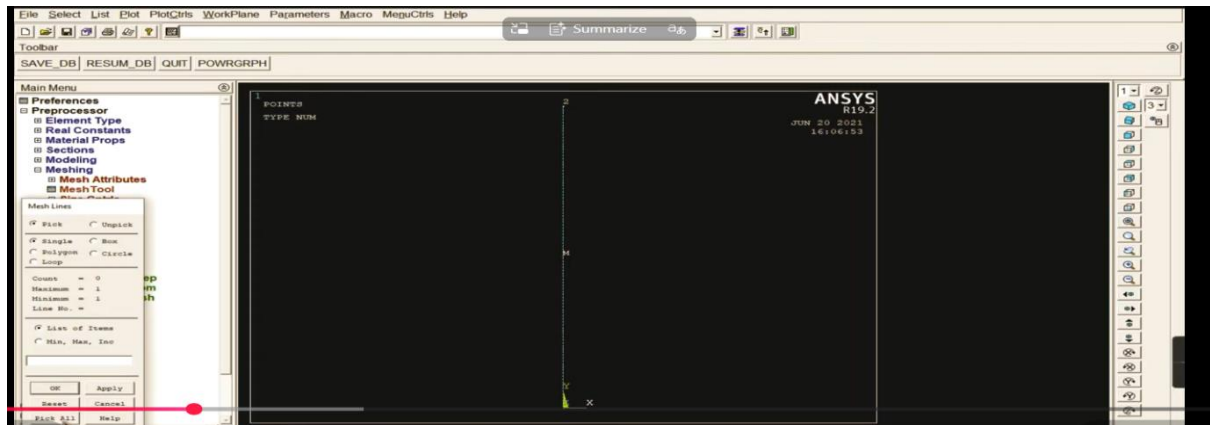


Step 5 Meshing

Meshing – Size Controls – Manual Size – Lines – All Lines – No of Element Division 100 - Ok

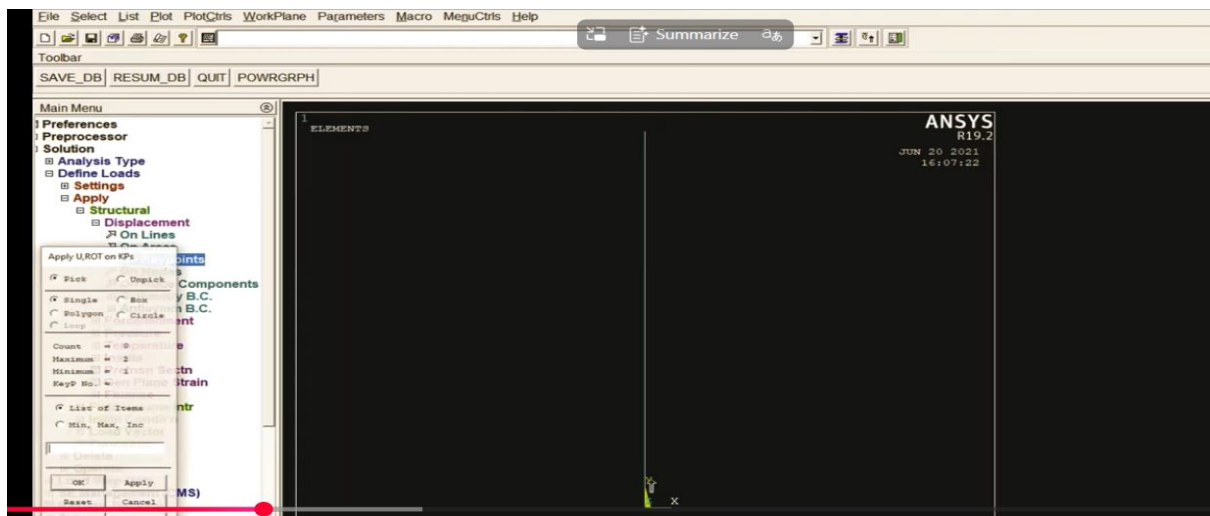
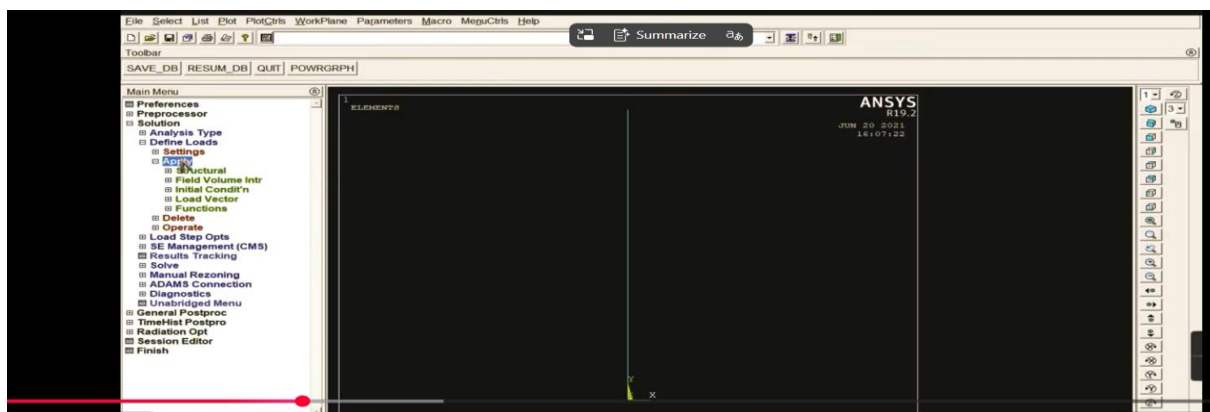


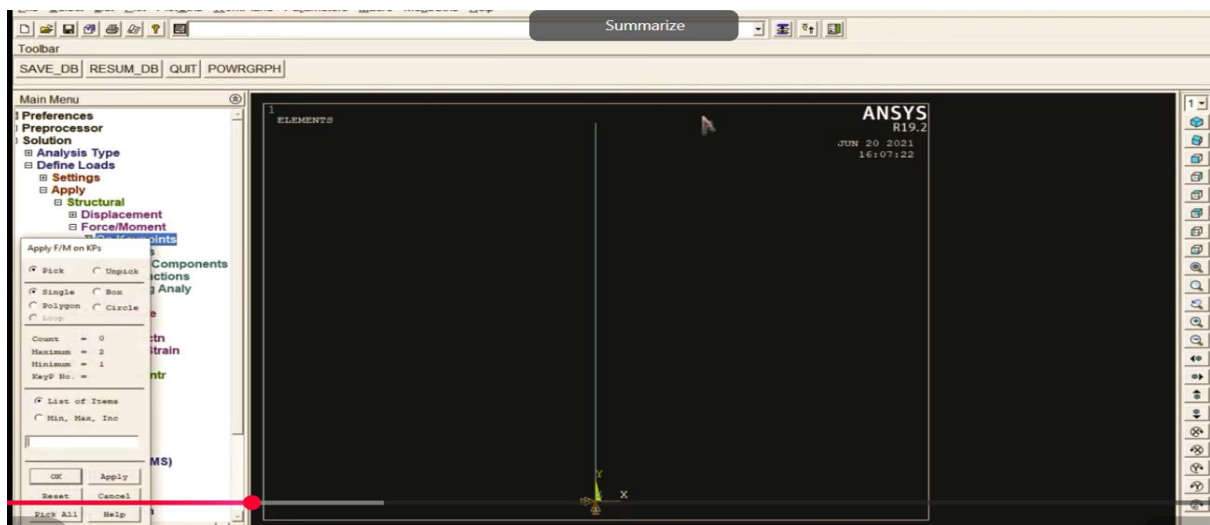
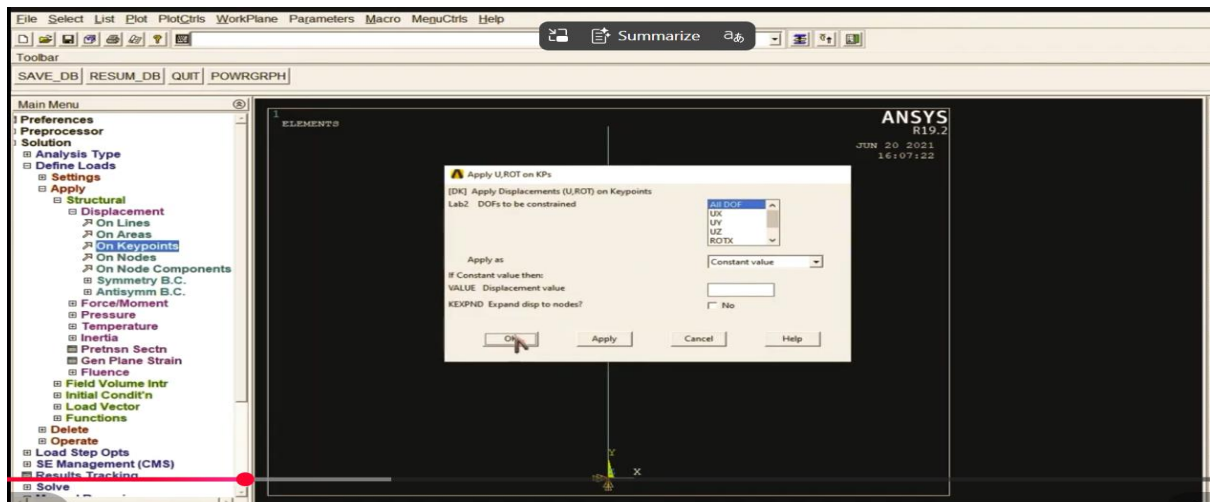
Mesh – Lines – Pick All



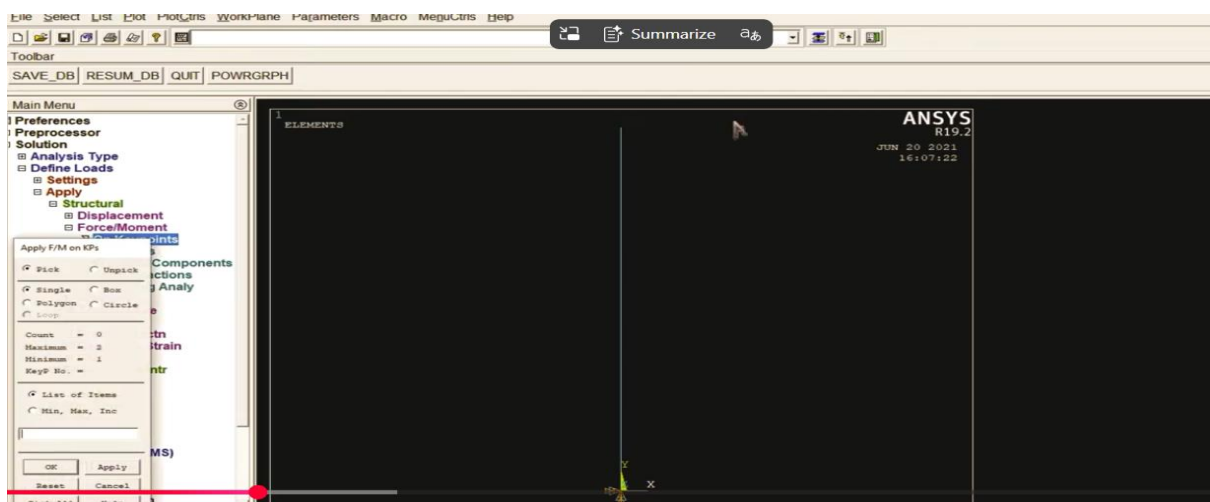
Step 6 Solution

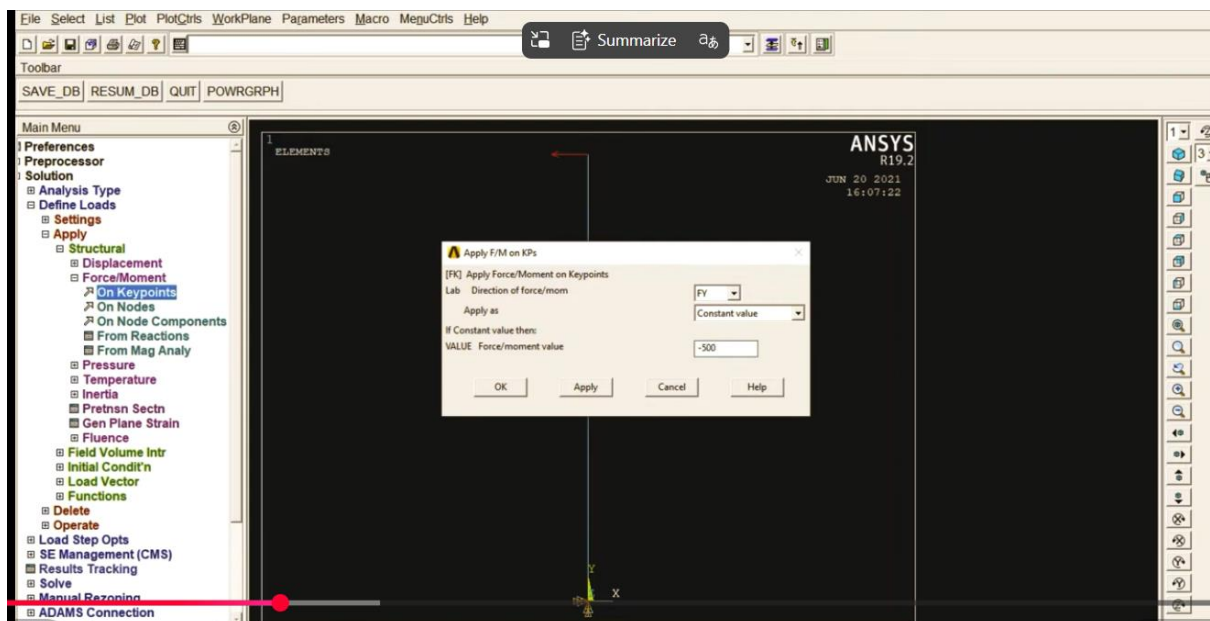
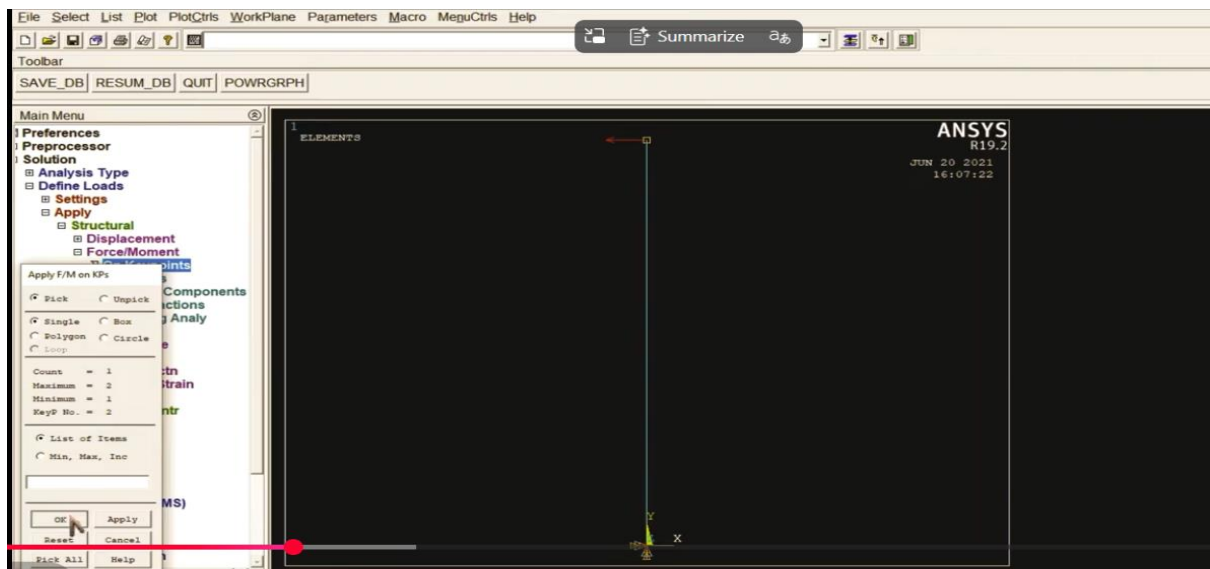
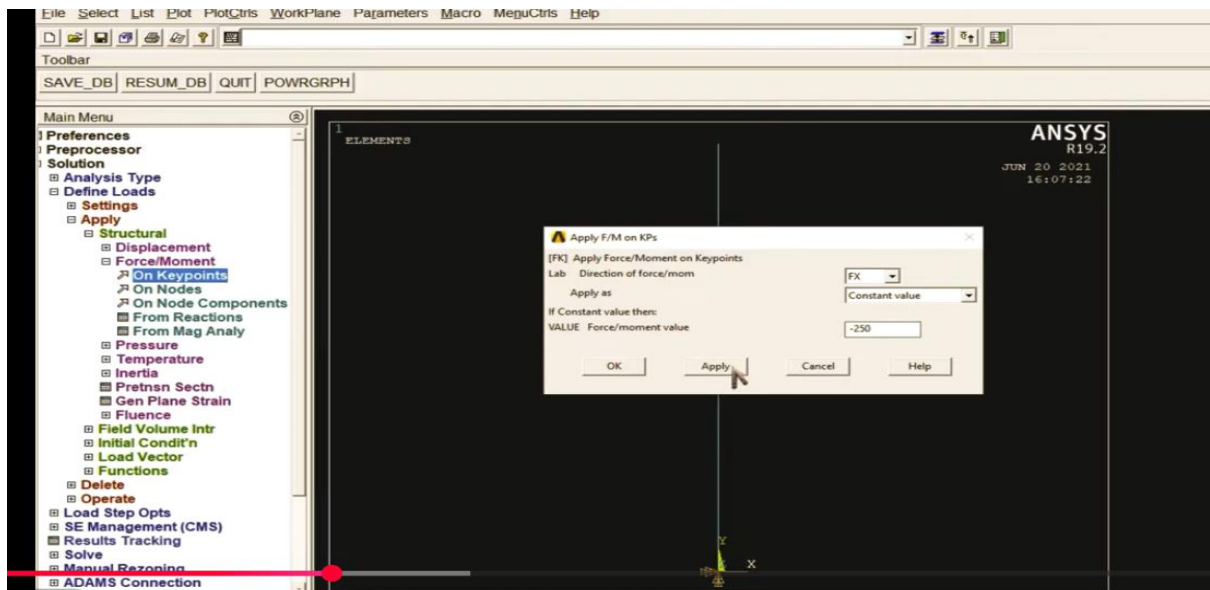
Loads – Define Loads –Apply – Structural – Displacement – Key points – Ok
– All DOF - Ok

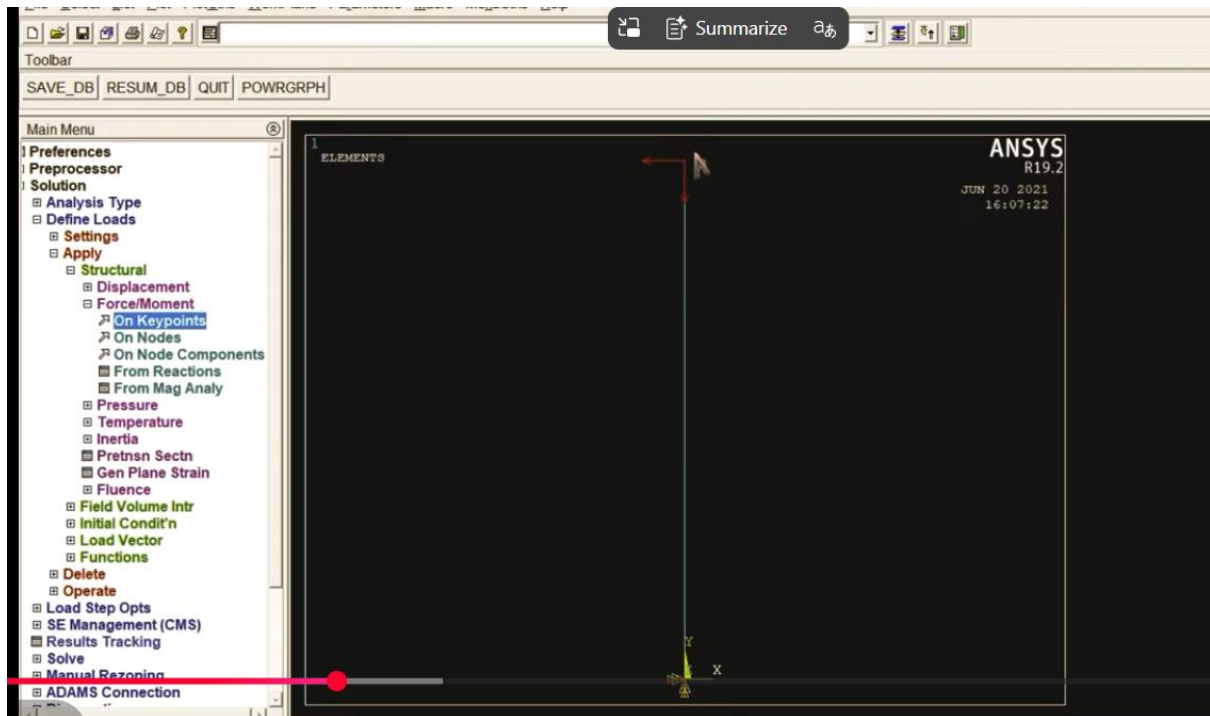




Loads – Define Loads –Apply – Structural – Force– Key points – FX - -250
– Apply – FY - -500 - Ok

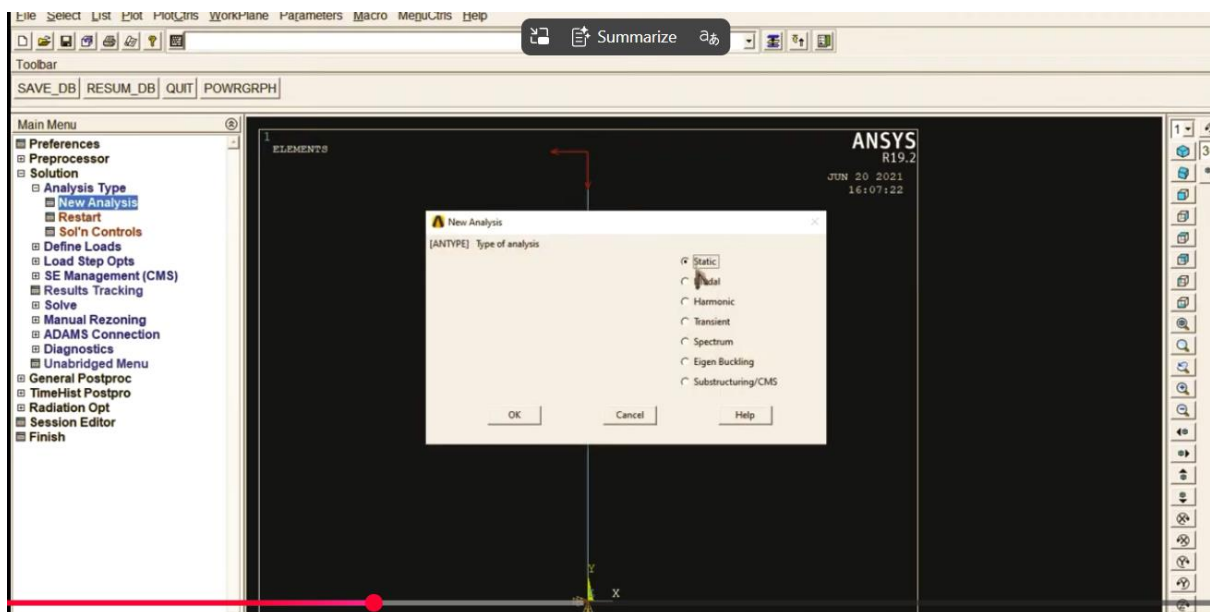






Step 7 Solution

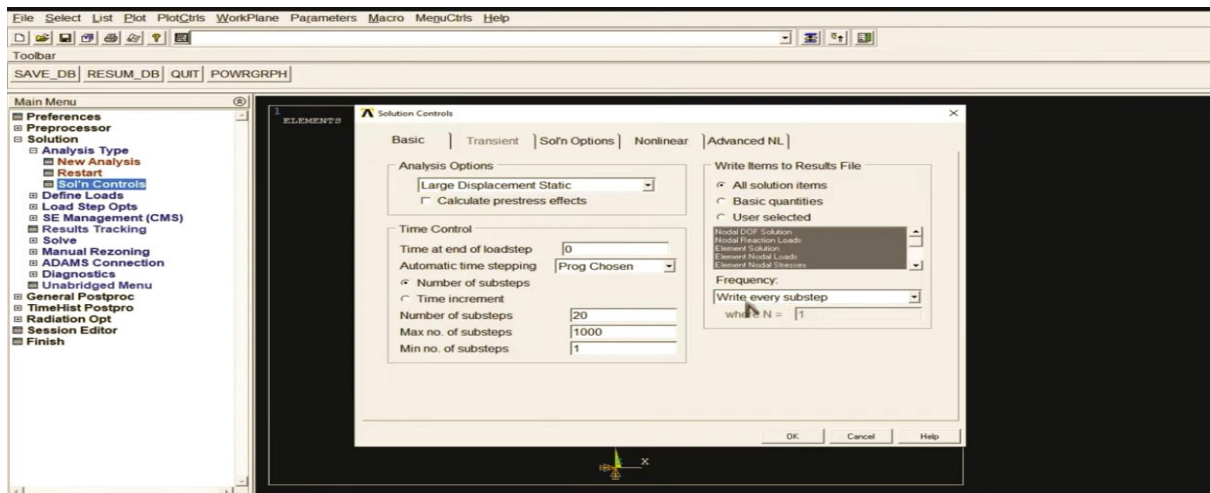
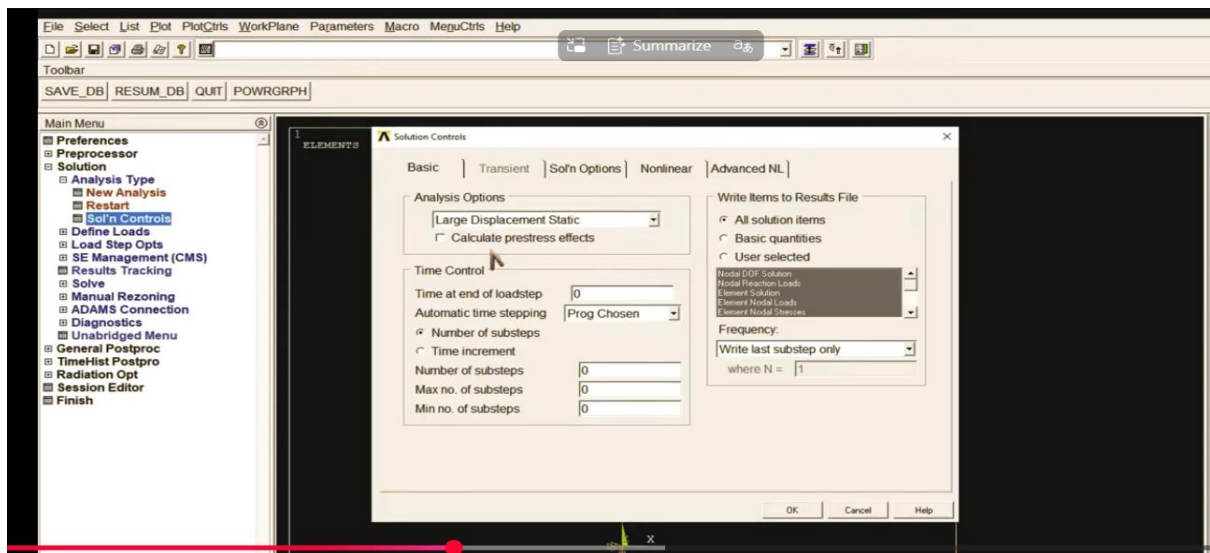
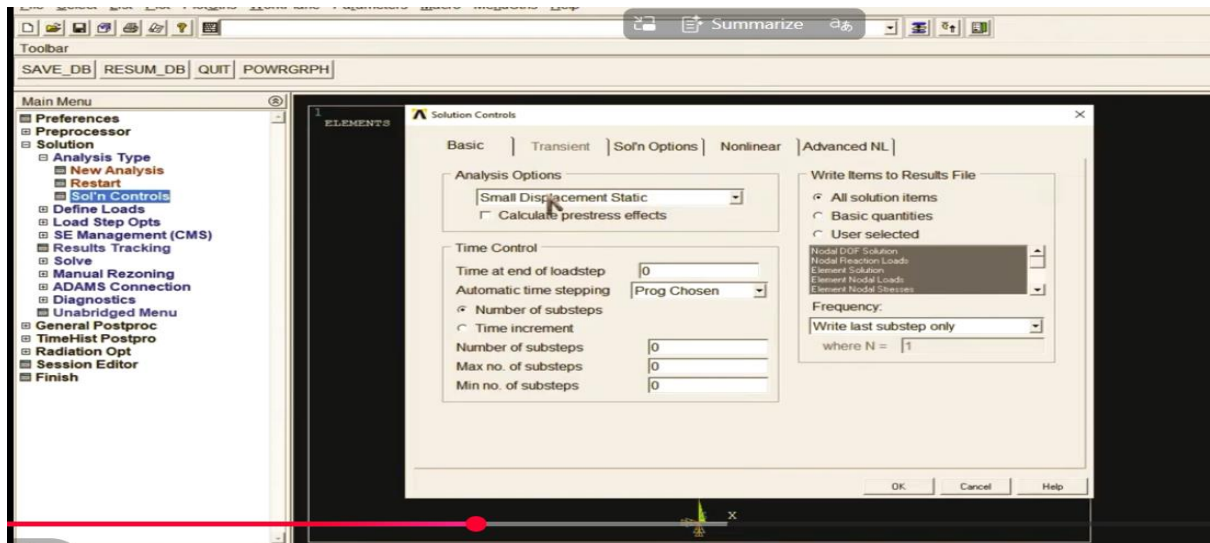
Solution – Analysis Type – New Analysis – Static - OK



Solution – Analysis Type – Solution Controls – Large Displacement Static –

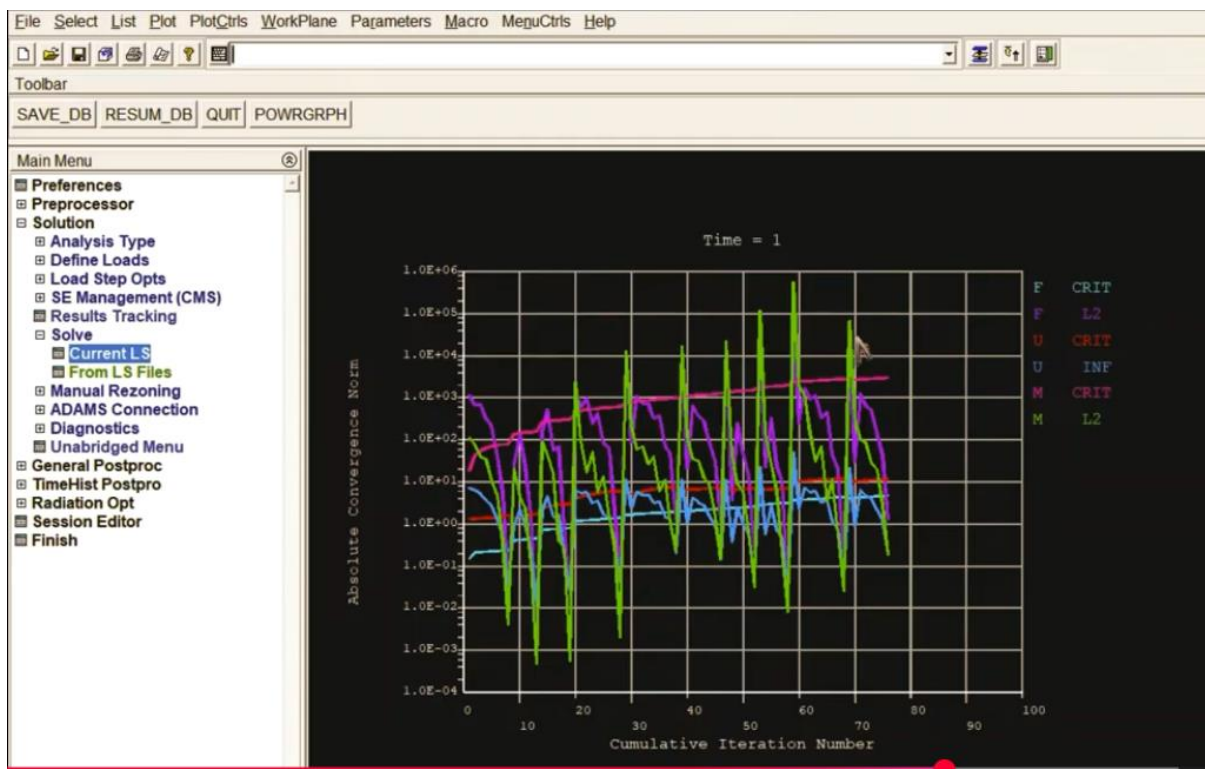
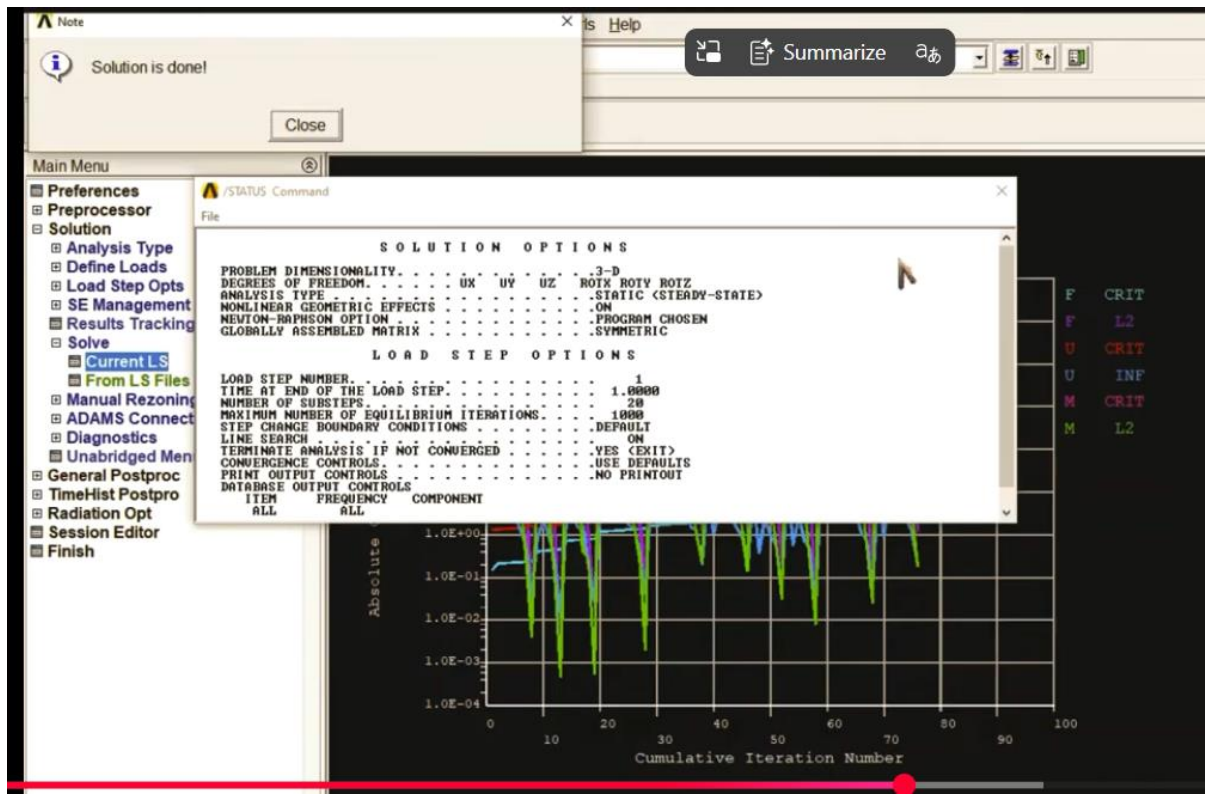
No of Sub steps 20 – Max number of sub step 1000 - min number of sub steps

Solution – Analysis Type – Solution Controls – Nonlinear – line search – on – maximum number of Iterations- 1000 - OK



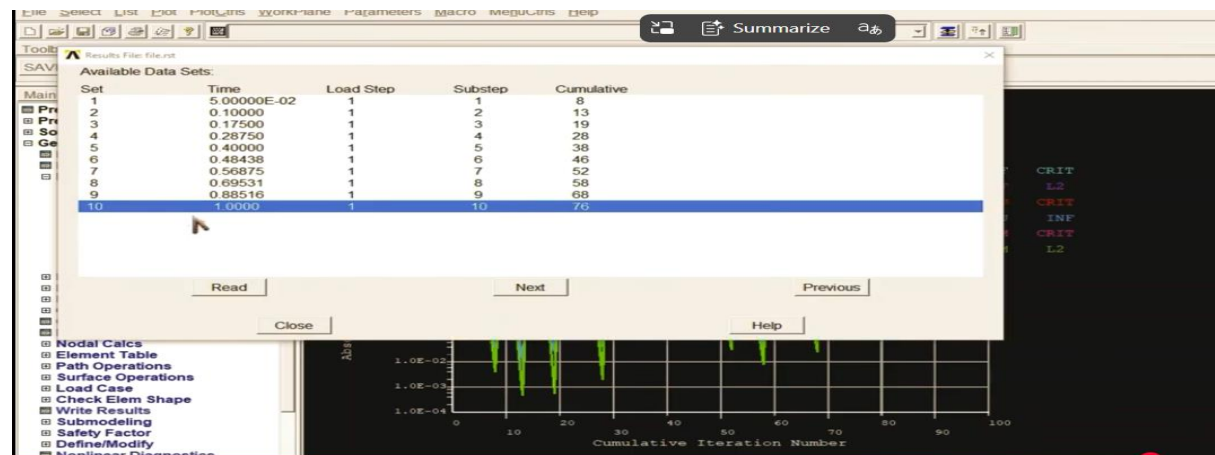
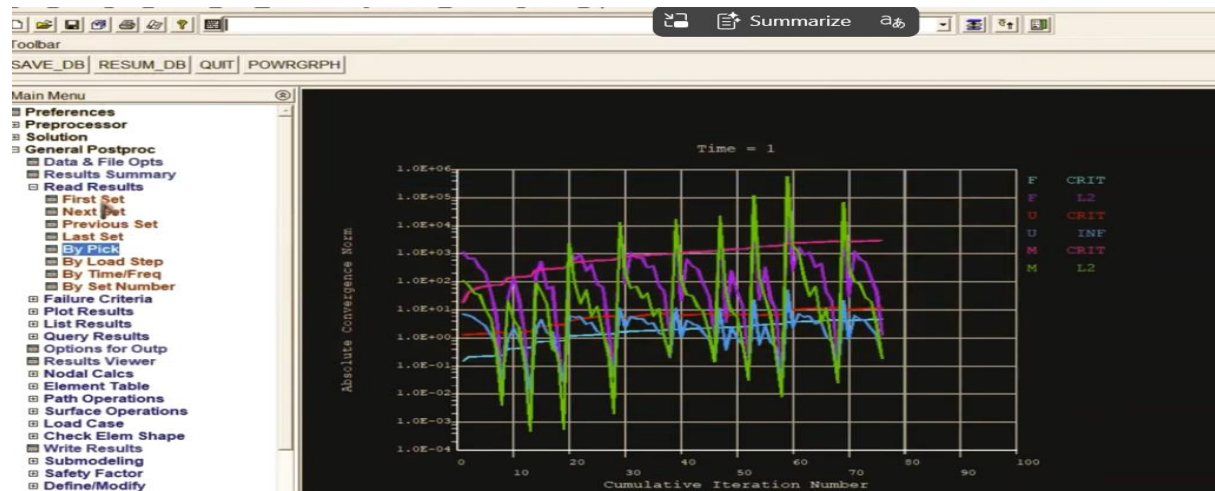
Step 8 Solution

Solution – Solve –Current LS - OK

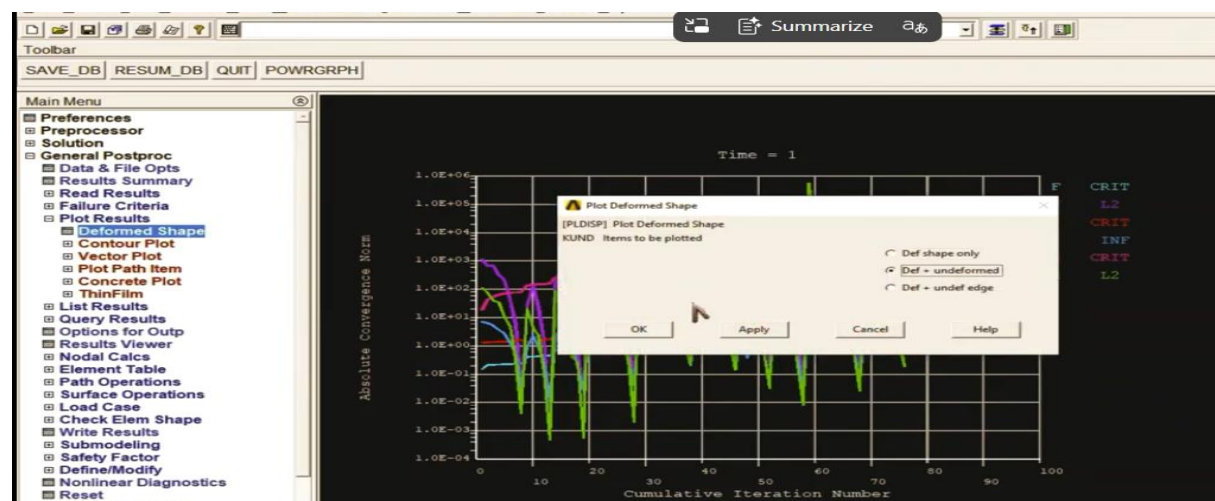


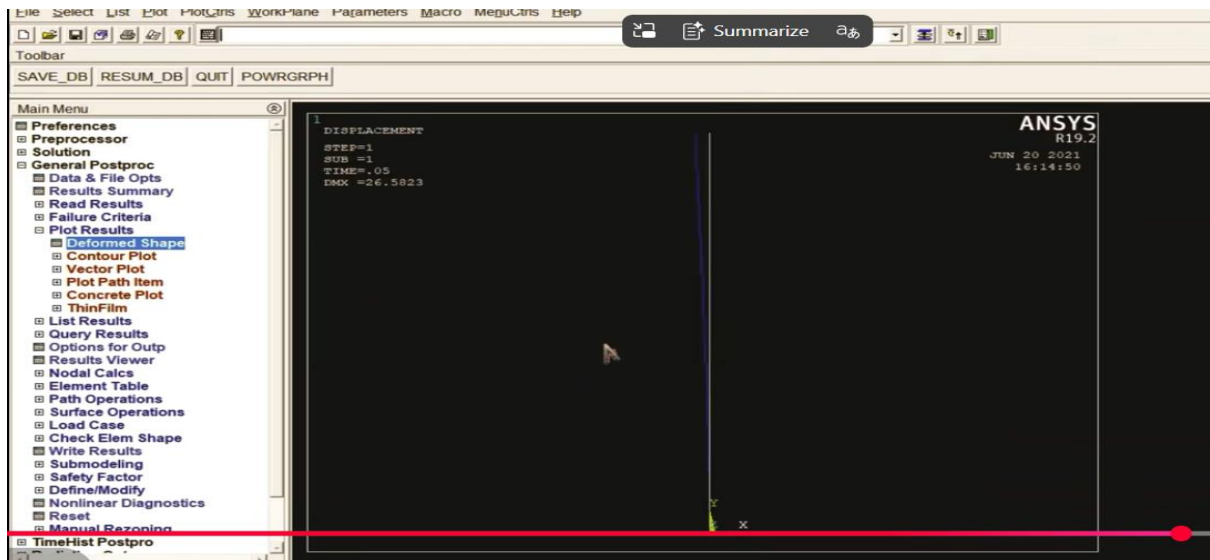
Step 9 General Results

General Post Process – Read Results – By Pic – Close

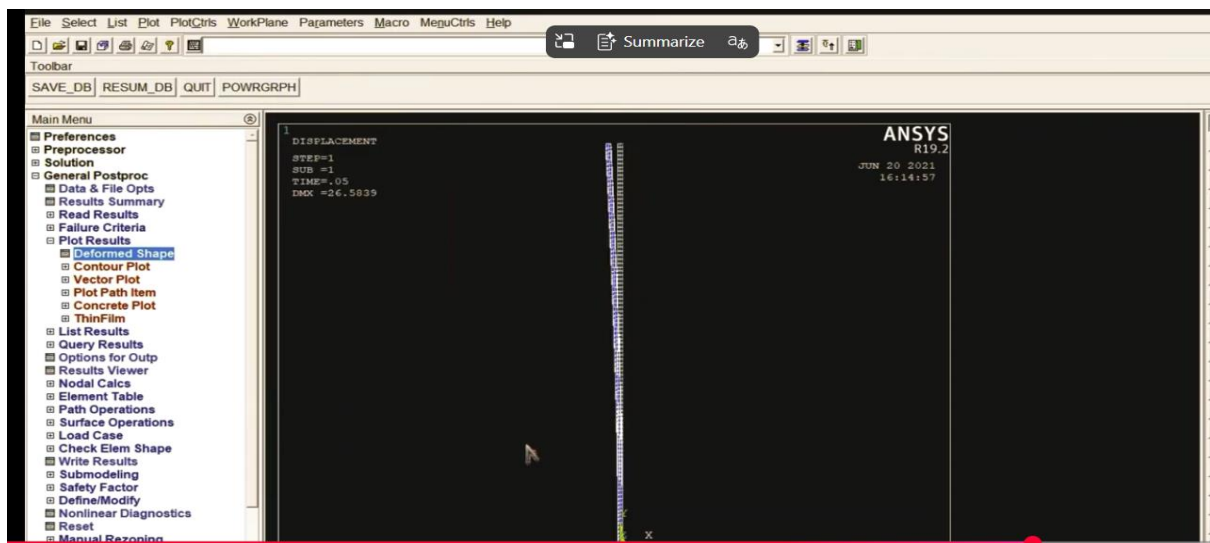
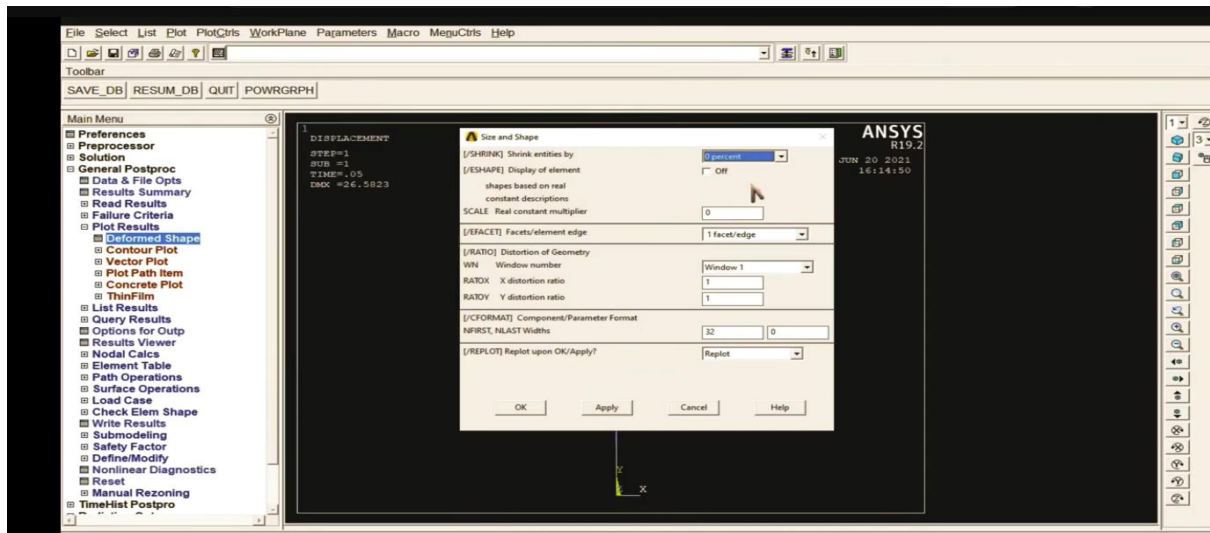


General Post Process – Plot Results – Deformed Shape – Deformed + Undeformed - OK

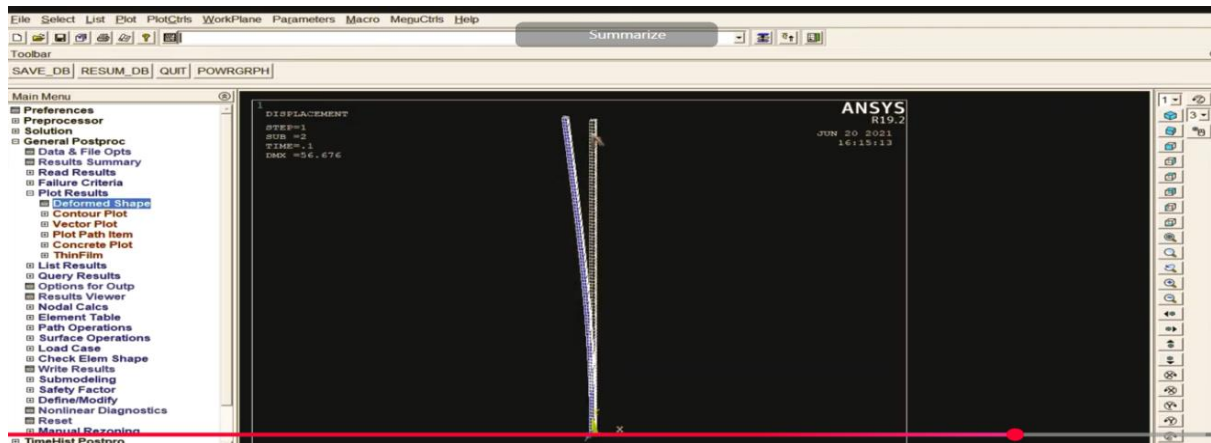




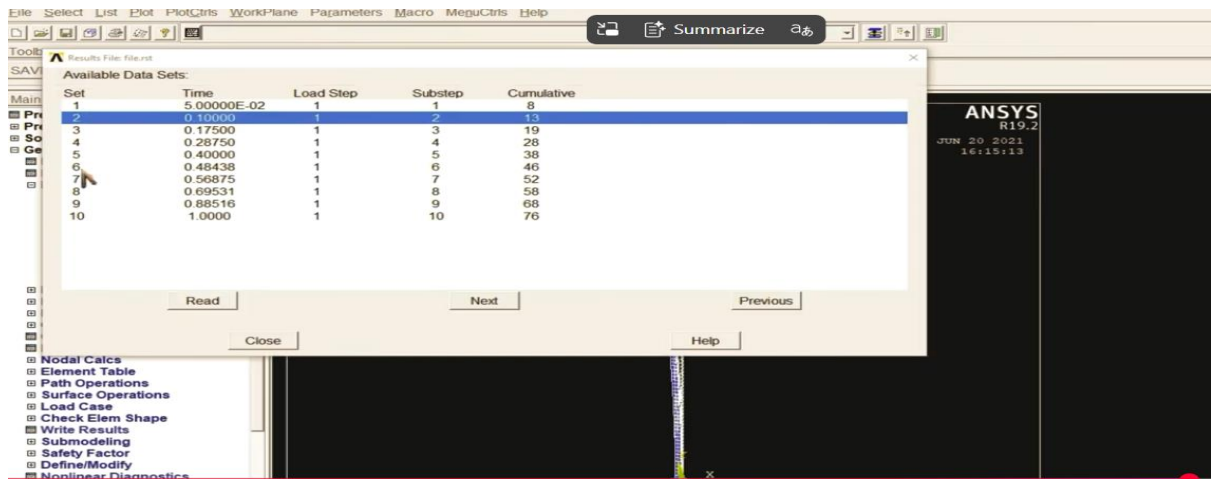
Plot Controls – Style – Size and Shape – Display of Shape - ON



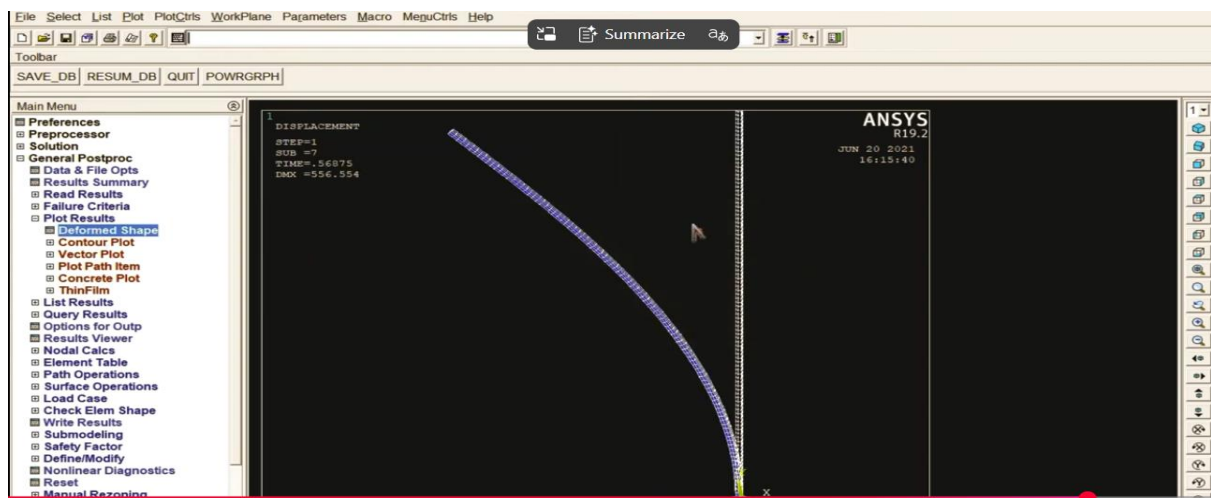
General Post Process – Plot Results – Deformed Shape -



General Post Process – Read Results – By Pick – Read - Close



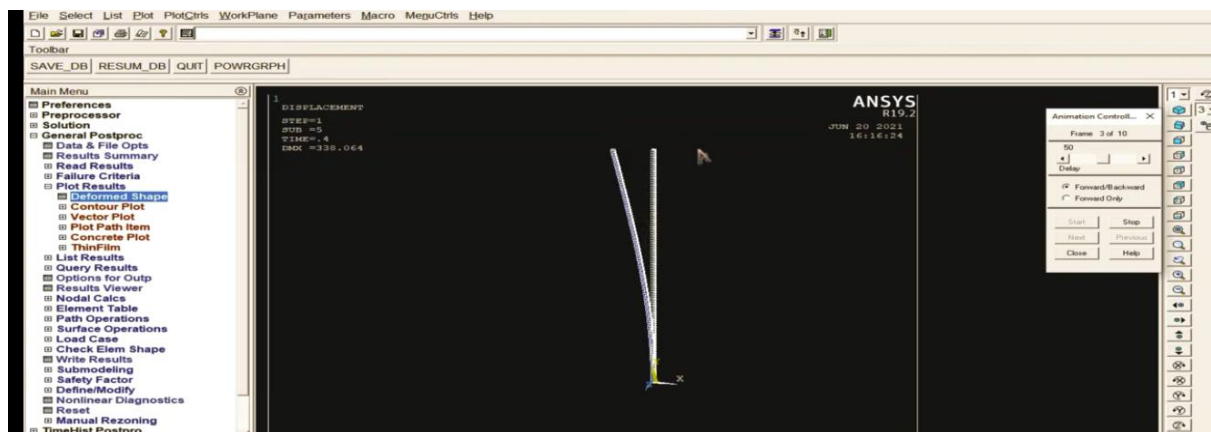
General Post Process – Plot Results – Deformed Shape -





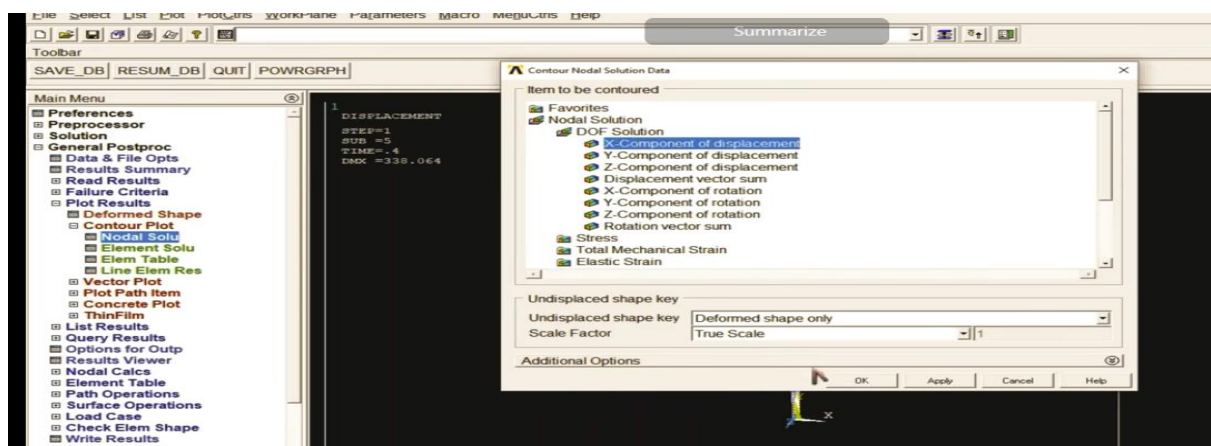
To See Animation

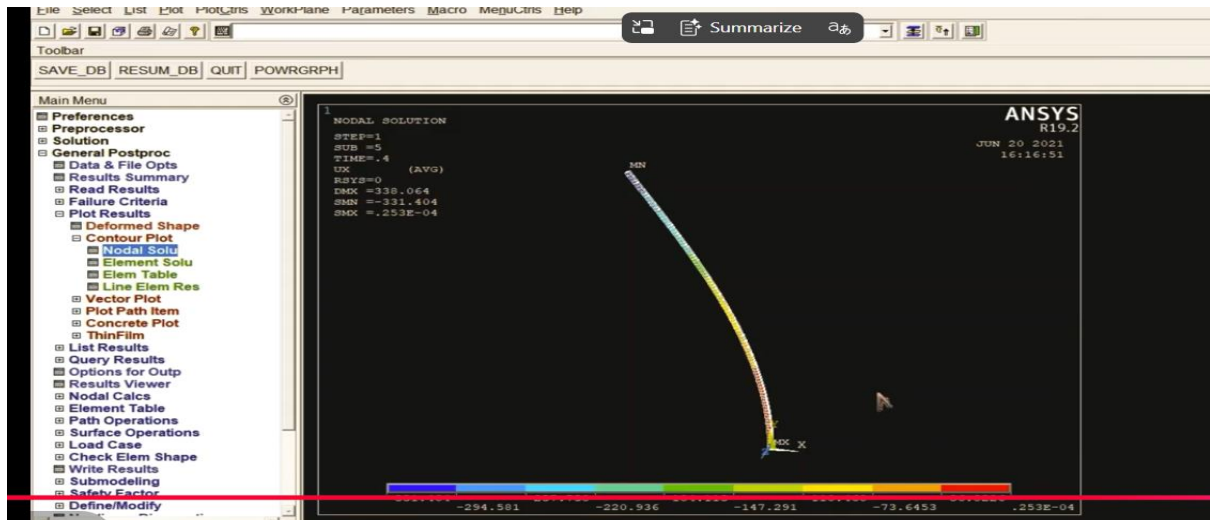
Plot Control – Animate – Deformed Shape – Deformed + Undeformed - OK



General Post Process – Plot Results – Contour Plot – Nodal Solution – DOF

Solution – X Component of displacement – ok





General Post Process – Plot Results – Contour Plot – Nodal Solution – DOF
Solution –displacement vector sum - ok

